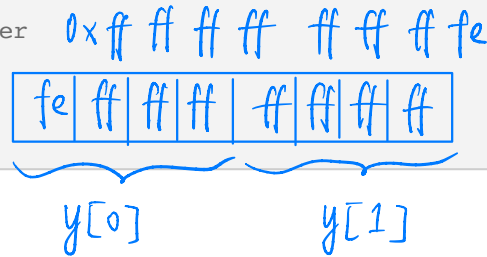


Q1 Pointer casting

1 Point

What's the output of the following code fragment (assuming it runs on a 64-bit little endian machine):

```
long long x = -2; //x is a signed 8-byte integer
int *y;
y = (int *)&x;
printf("%d %d\n", y[0], y[1]);
```



-1 -1

-2 -2

-1 -2

-2 -1

Segmentation fault

None of the above

Q2 ASCII

1 Point

```
#include <stdio.h>
int main() {
    char c = '0';
    printf("%d %c\n", c, c + 1);
}
```

What does the above program print?

0 1

0 0

48 49

48 0

48 1

Q3 String

1 Point

Below is the `strlen` function that returns a C string's length (it's also a `stdlib` function):

```
int strlen(char *s) {  
    int r = 0;  
    while ((*s)!='\0') {  
        r++;  
        s++;  
    }  
    return r;  
}
```

```
1: char *s = "hello";  
2: int x = strlen(s);
```

string literal: implicit '\0' in the end at compile time

What is the value of x after executing the above line?

Segmentation Fault

5

6

0

x's value is undefined (i.e. could be any int value).

Q4 String

1 Point


1: `char *s = "cso";`
2: `int x = strlen(s+1);`

What's the value of `x` after the above two lines of code?

Compilation error at line 1

Compilation error at line 2

`x = 0`

`x = 1`

`x = 2`

`x = 3`

`x = 4`

`x`'s value is undefined (i.e. could be any int value).

Q5 String

1 Point

```
1: int a = 0x00414243;  
2: int x = strlen((char *)&a);
```

43	42	41	00
----	----	----	----

What's the value of `x` after the above two lines of code?

Compilation error at line 1

Compilation error at line 2

`x = 0`

`x = 1`

`x = 2`

`x = 3`

`x`'s value is undefined (i.e. could be any int value).

Q6

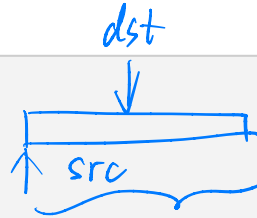
3 Points

Q6.1

1 Point

Below is the implementation of `my_strncpy` which copies $\min(n, \text{strlen}(\text{src})+1)$ bytes from string `src` to string `dst`.

```
void my_strncpy(char *dst, char *src, int n)
{
    for (int i = 0; i < n; i++) {
        dst[i] = src[i];
        if ((src[i]) == '\0') return;
    }
}
```



need to iterate to '\0'

```
1: char d[____];
2: my_strncpy(d, "hi", ____);
3: printf("%s\n", d);
```



What of the following numbers can be used to filled in the blank in line 1 and line 2 respectively such that the above code correctly prints out "hi" for all execution?

2 2

2 3

3 2

3 3

Q6.2

1 Point

```
1: char d[6] = {'h', 'e', 'l', 'l', 'o'};  
2: my_strncpy(d+2, "hi", 3);  
3: printf("%s\n", d);
```

Handwritten annotations:
- Above line 1: 'h', 'e', 'l', 'l', 'o' with single quotes and arrows pointing to the corresponding characters in the array.
- Next to line 2: 'd+2' with an arrow pointing to the third character 'l' in the array.

What does the above code print out?

hello

hhilo

hehio

hehi

heh

hehillo

Q6.3

1 Point

Below is a function that concatenates two strings and returns the combined result as a new string:

```
1: char *  
2: str_concat(char *s1, char *s2)  
3: {  
4: _____; //define s and allocates space  
5: my_strncpy(s, s1, strlen(s1));  
6: my_strncpy(s+strlen(s1), s2, strlen(s2)+1);  
7: return s;  
8: }
```

For a correct implementation of `str_concat`, what should be on line 4:

`char *s = s1;`

`char *s = s2;`

`char s[strlen(s1)+strlen(s2)];`

`char s[strlen(s1)+strlen(s2)+1];`

`char *s = (char *)malloc(strlen(s1)+strlen(s2));`

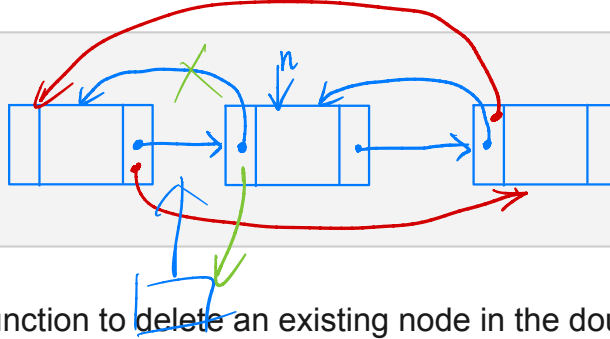
`char *s = (char *)malloc(strlen(s1)+strlen(s2)+1);`

Q7 Doubly linked list

2 Points

Suppose a doubly linked list has the following type definition for its nodes:

```
typedef struct lnode {  
    int val;  
    struct lnode *next;  
    struct lnode *prev;  
}lnode_t;
```



Ben Bitdiddle writes the following function to delete an existing node in the doubly linked list:

```
//n is a node in the doubly linked list whose head node is pointed to by *headdp  
//Unlink n from the list  
1: void  
2: delete_node(lnode_t **headdp, lnode_t *n) {  
  
3:     //link n's predecessor (if one exists) to n's successor  
4:     if (n->prev) _____  
  
5:     //link n's successor (if one exists) to n's predecessor  
6:     if (n->next) _____  
  
7:     //if n is the first node in list, change head pointer to point to n's successor.  
8:     if (n->prev == NULL) *headdp = n->next;  
  
9:     free(n); //de-allocate n  
8: }
```

Q7.1

1 Point

Please fill in the _____ at line 4 (This is supposed to be a oneliner. To ensure automatic grading, please terminate your oneliner with `;` and do not have any space characters inside)

```
n->prev->next=n->next;
```


Q7.2**1 Point**

Please fill in the ____ at line 6 (This is supposed to be a oneliner. To ensure automatic grading, please terminate your onliner with ; and do not have any space characters inside)

```
n->next->prev=n->prev;
```